



HYDROLOGICAL BALANCE OF THE NIGER BASIN



YEAR 2024/2025

Table of contents

1.	ANALYSIS OF THE EVOLUTION OF FLOW FROM THE NIGER BASIN	5
1.1	Upper Niger	5
1.2	Inner Niger Delta	6
1.3	Middle Niger	7
1.4	Lower Niger	7
2.	SITUATION OF WATER RETAILERS IN DAMS	8
2.1	The Sélingué Dam at Ma li	8
2.2	The Kainji Dam in Nigeria	9
	CONCLUSION	10

INTRODUCTION

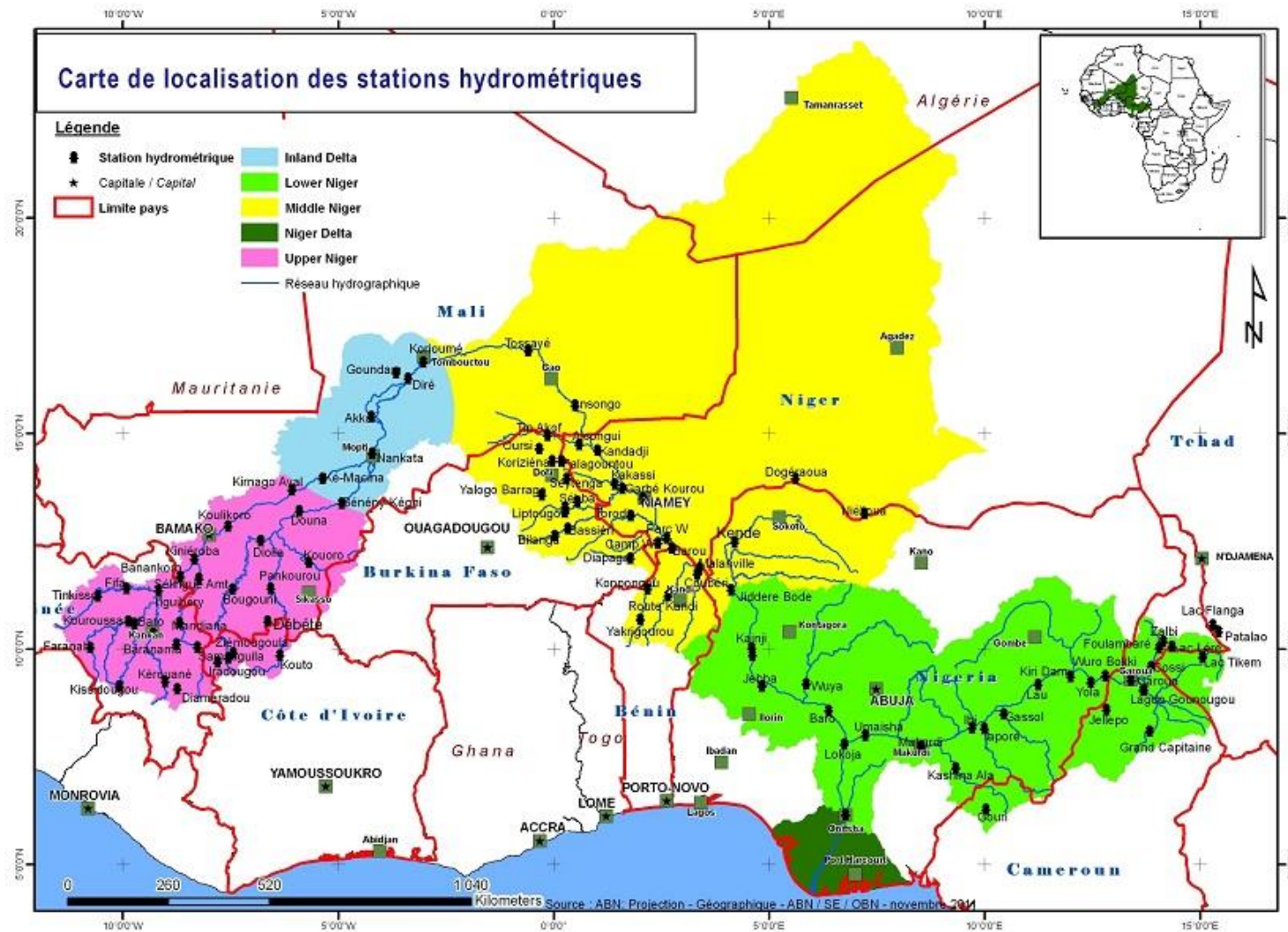
The Niger River takes its source in the plateaus of the Fouta-Djallon at an altitude of more than 1,000 m in Guinea and successively crosses Mali, Niger, Benin and Nigeria where it flows into the Atlantic Ocean after a journey of approximately 4,200 km. It is the third longest river in Africa (after the Nile and the Congo), the 14thⁱⁿ the world and the 9thⁱⁿ terms of its drainage basin (2,170,500 km²). Its watershed active covers an area of approximately 1,500,000 km² shared by the nine (9) countries grouped within the Niger Basin Authority: Benin (2%), Burkina Faso (4%), Cameroon (4%), Côte d'Ivoire (1%), Guinea (6%), Mali (26%), Niger (23%), Nigeria (33%) and Chad (1%). The Benue which has its source in Cameroon, receives contributions from the Mayo Kebbi River from Chad, before joining the main course of the Niger River at the confluence of Lokoja in Nigeria, is its most important tributary.

The hydrological year of the Niger River goes from June 1st of a year to May 31st of the following year.

Due to differences between physiographic and climatic characteristics, the Niger River basin is subdivided into 4 sub-basins: The Upper Niger, the Inner Delta, the Middle Niger and the Lower Niger. This annual report of the Niger River for the hydrological year 2024/2025 analyzes the flow situation at stations representative of the different sub-basins: Koulikoro (Mali) for the Upper Niger, Diré (Mali) for the Inner Delta; Niamey (Niger) for Middle Niger and Lokoja (Nigeria) for Lower Niger (See Fig.1.a)

The flow monitoring during the hydrological year 2024/2025 was carried out as part of the activities of the Niger Basin Observatory from a hundred hydrometric stations in the hydrological observation networks of the National Hydrological Services and Dam Agencies of NBA member countries. These stations are equipped with Satlink type data collection platforms (PCD), electronic limnigraphs with or without GSM transmission, or simple limnimetric scales with an observer agent who ensures daily readings.

Fig.1.a Location map of hydrometric stations in the Niger Basin



1. ANALYSIS OF FLOW EVOLUTION FROM THE NIGER BASIN

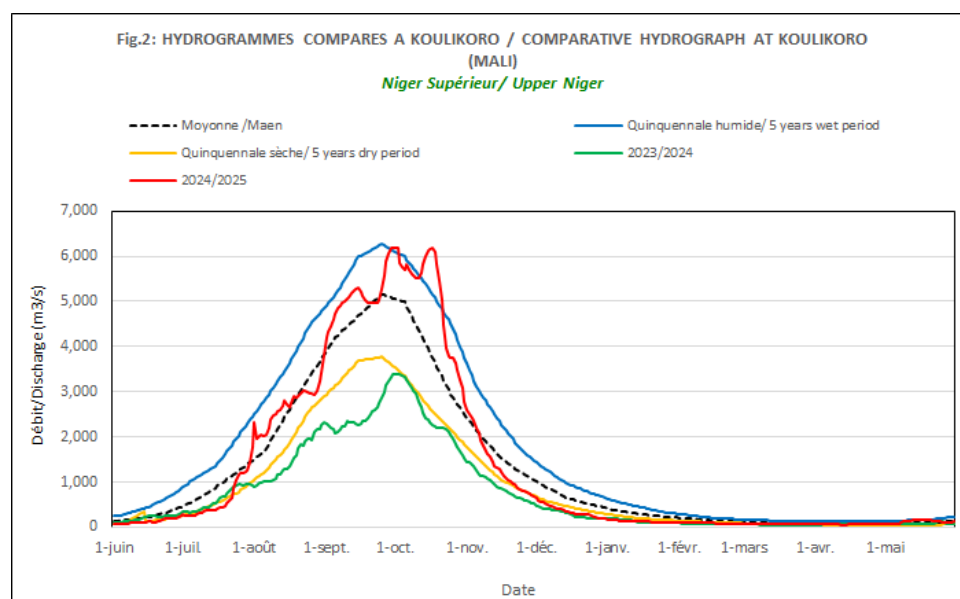
As indicated above, the analysis of the hydrological situation concerns the hydrometric stations of the four sub-watersheds represented by: Koulikoro (Mali) for the Upper Niger, Diré (Mali) for the Inner Delta; Niamey (Niger) for Middle Niger and Lokoja (Nigeria) for Lower Niger.

Figures 2 to 5 present the hydrographs compared from June to May for the hydrological years of 2024/2025, 2023/2024, the five-years dry and wet periods and the inter-annual mean of flows. Figures 6 and 7 respectively illustrate the annual water level of the Sélingué dams in Mali and the Kainji dams in Nigeria. Table 1 respectively summarizes the cumulative water volumes during the hydrological year 2024/2025 at the reference stations in the Niger basin.

1.1 Upper Niger

In Koulikoro in Mali in Upper Niger, the volume of water flowed from June 1st, 2024 to May 31st, 2025 was approximately 42.0 billion m³. The average flow at the Koulikoro station during the hydrological year 2024/2025 is 1,331 m³/s. The daily maximum of 6192 m³/s was observed on October 17th, 2024 and the daily minimum of 55 m³/s observed on April 12th, 2025.

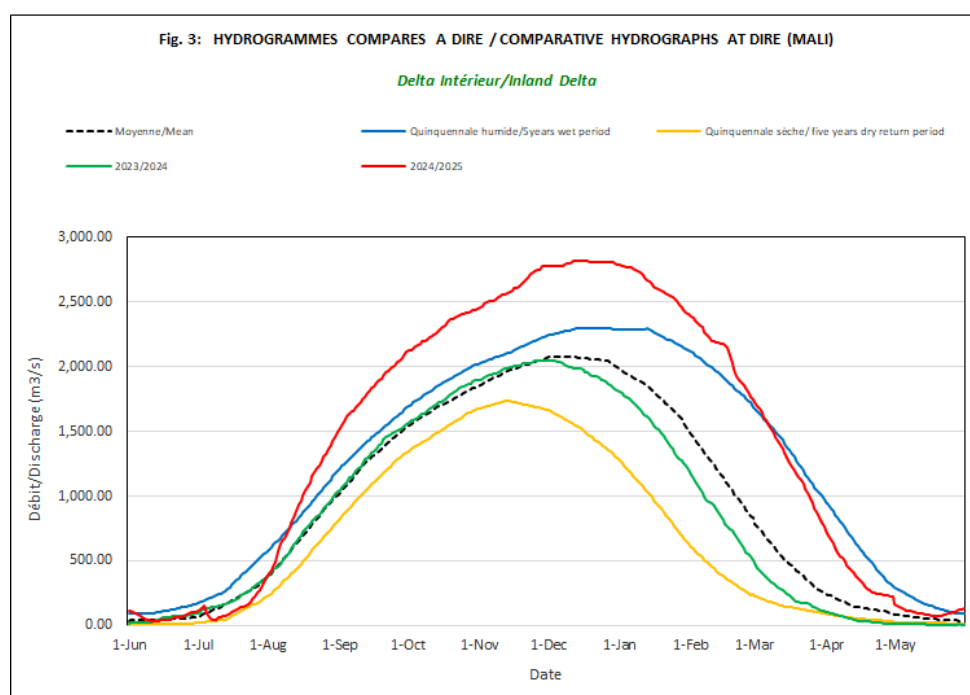
The hydrological year 2024/2025 at the Koulikoro station in Upper Niger was normal with a total flow more than that of the year 2023/2024, the inter-annual average, the five-year dry period but lower than the five- years wet period.



1.2 Inner Niger Delta

In the Inner Niger Delta (DIN), at the Diré station, the volume of water flowed from June 1st, 2024 to May 31st, 2025 was approximately 45.20 billion m³. The average flow rate during the hydrological year 2024/2025 at DIN station was 1432 m³/s. The maximum daily flow of 2820 m³/s was recorded on December 12th, 2024 and the minimum daily flow of 36 m³/s was recorded on June 13th, 2024.

The flow rates at the Diré station for the hydrological year 2024/2025 was higher than all the years of comparison: the year 2023/2024, five-years wet values, the five-years dry values and the inter-annual average during the same period. The 2024/2025 water year in the DIN was a wet year due to the increased flow and rainfall recorded in the sub-basin.

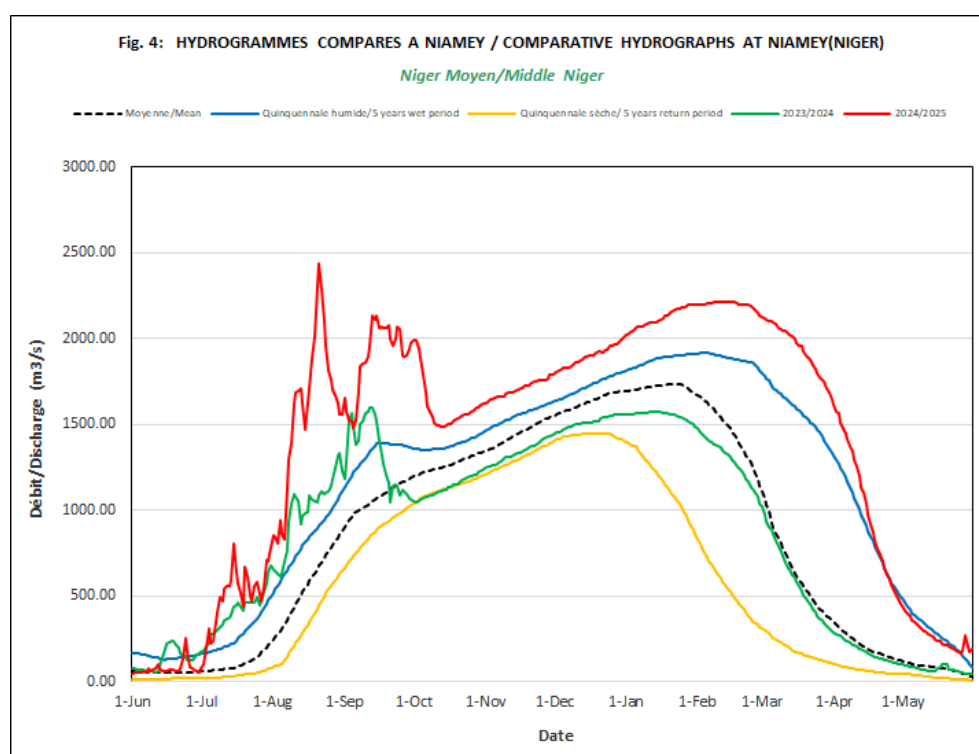


1.3 Middle Niger

In the Middle Niger in Niamey, Niger, the volume of water flowed from June 1st, 2024 to May 31st, 2025 was approximately 43.97 billion m³. The average flow during the hydrological year 2024/2025 was 1394 m³/s. A maximum daily flow of 2438 m³/s was recorded on August 21st, 2024 and a minimum daily flow of 49 m³/s recorded on June 1st, 2024.

The average flow rates for this hydrological year 2024/2025 was higher than all the years of comparison: the year 2023/2024, five-years wet values, the five-years dry values and the inter-annual average during the same period.

The hydrological year 2024/2025 at the middle Niger was a wet year due to the contributions from the tributaries of the right bank (Dargol, Gorouol, Sirba, Goroubi) which take their source from Burkina Faso and the Niger River course coming from the Upper sub-basin.

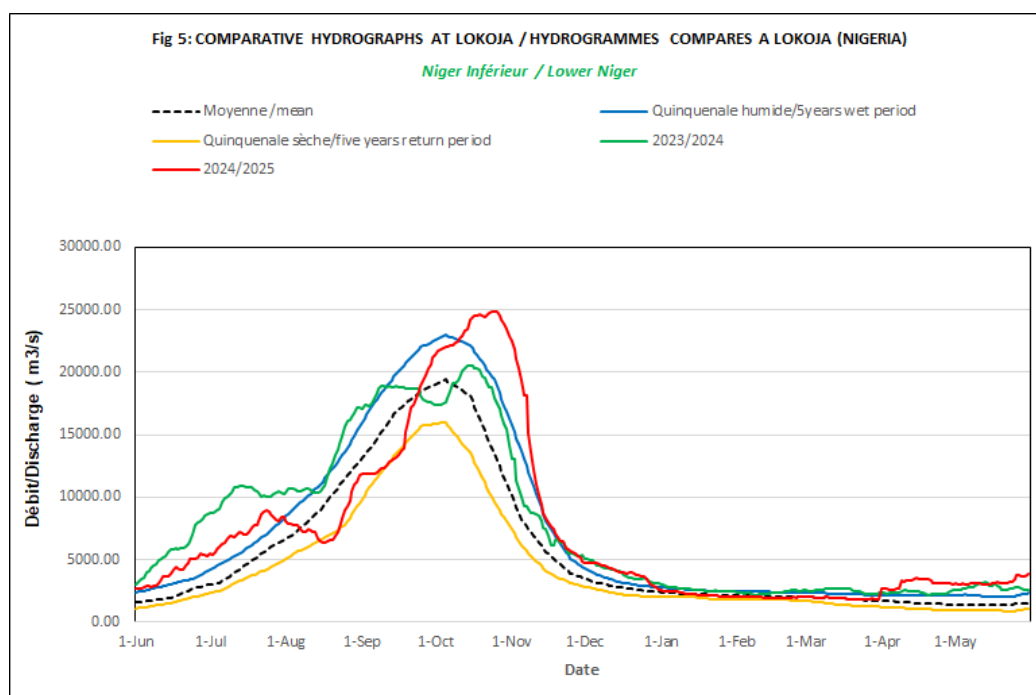


1.4 Lower Niger

In the Lower Niger at Lokoja, Nigeria, the volume of water flowed from June 1st, 2024 to May 31st, 2025 was approximately 224 billion m³. The average flow during the hydrological year 2024/2025 recorded at the Lokoja station was 7100 m³/s. A maximum daily flow of 24835 m³/s recorded on October 24th, 2024 and a minimum daily flow of 1826 m³/s recorded on March 23rd, 2025.

The average flow rates recorded at this station during this hydrological year are higher than those of the inter-annual average (1981-2010) and the five years dry values but lower than those of the years of the 2023/2024 and the five years wet values during the same period.

The hydrological year 2024/2025 in the Lower Niger was a wet year due to contributions from the Benue tributary and those from the Niger River artery.



2. SITUATION OF WATER RETAINED IN DAMS

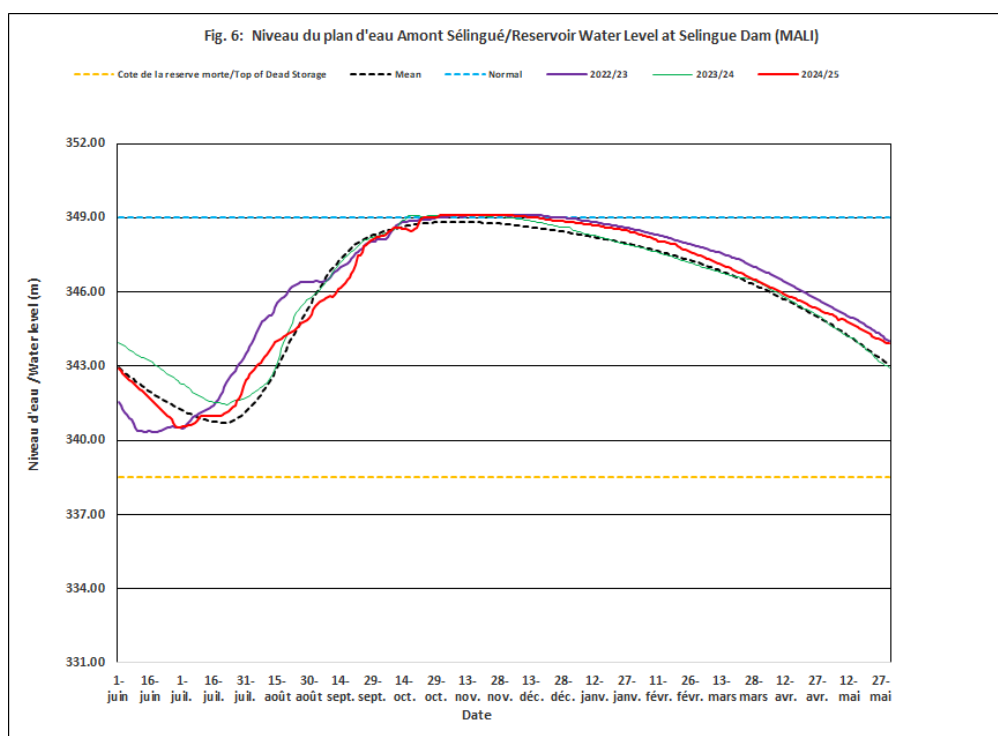
2.1 The Sélingué Dam in Mali

The annual maximum water level of 349.13 m was recorded from November 1st and 3rd, 2024. The annual minimum level of 340.52 m was recorded on June 29, 2024 with an inter-annual average of 346.16 m. This average level remains lower than that of the years 2023/2024, 2022/2023 but higher than that of the mean average values as shown in Figure 6.

The normal dam level of 349.00 m was reached on October 22nd, 2024 while the spill began on October 23rd and continued until December 15th, 2024.

Emptying of the reservoir began on December 17th, 2024 (348.99 m) and continued until May 31st, 2025 (343.91 m) to support low water levels downstream.

It should be noted that the only major flow regulation structure in the upper Niger is the Sélingué dam, built on the Sankarani, a tributary of the Niger River. This dam has a capacity of 2.347 billion m³ and makes it possible to produce electricity for an installed capacity of 47.6 MW on average and to support low flow rates to ensure rice production and improve navigation downstream of the dam.



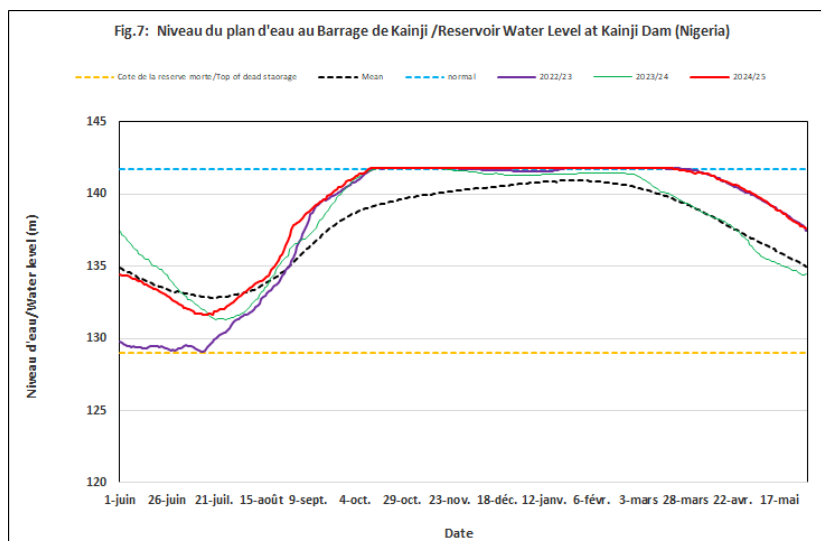
2.2 Kainji Dam in Nigeria

The annual maximum water level of 141.83 m was recorded from October 12th 2024 to March 17th, 2025. The annual minimum water level of 131.65 m was recorded on June 16th and 17th, 2024 with the inter- annual water level average of 139.16m. This water level was higher than all the years of comparison: the year 2023/2024, the year 2022/2023 and the annual average value as shown in Figure 7.

The normal water level of 141.73 m was reached on October 11th, 2024 and the spill began on October 12th, 2024 and continued until March 22nd, 2025.

Emptying of the reservoir began on March 24th, 2025 (141.72 m) and continued until May 31st, 2025 (137.65 m) to support low water levels downstream.

This large capacity dam of 15 billion m³ normal reservoir, is intended for hydroelectric production with an installed capacity of 720 MW. It plays an important role in regulating flows from the Upper Niger and the Middle Niger. Also, the hydroelectric production from the Jebba dam located downstream is inter-dependent on releases from the Kainji dam for maximum power generation.



CONCLUSION

Generally speaking, the hydrological year 2024/2025 was a wet year with flows more than the inter-annual average values in all the sub-basins. This, among other things were as a result of more intensity of rainfall in the basin.

As for the water levels of the Sélingué and Kainji dams during the hydrological year 2024/2025, they reached their normal level of 349.00 m on October 22nd, 2024 and 141.73 m on October 11th, 2024 respectively. Also, throughout the year, these hydraulic structures supported low water levels downstream in order to meet the different needs of users.

The Hydrological situation along the Niger River can be consulted on the following websites: www.abn.ne; <http://nigerhycos.abn.ne>; www.sath.abn.ne. For all comments and suggestions, please send them to one of the following emails: bamidele.olatunji@bassin-niger.org; b.coulibaly@bassin-niger.org; bachir.tanimoun@bassin-niger.org

Table 1: Cumulative water volumes from June 1st 2024 to May 31st 2025 at the reference stations.

STATIONS	YEARS/YEAR	CUM (10⁹ m3)
UPPER NIGER / UPPER NIGER (KOULIKORO)	2024/25	42.00
	2023/24	23.60
	Five-year wet period	55.65
	Five-year dry period	28.78
	Mean	40.43
INTERIOR DELTA/INNER DELTA (SAY)	2024/25	45.20
	2023/24	28.02
	Five-year wet period	40.22
	Five-year dry period	21.12
	Mean	30.53
MIDDLE NIGER / MIDDLE NIGER (NIAMEY)	2024/25	43.97
	2023/24	27.56
	Five-year wet period	35.94
	Five-year dry period	18.72
	Mean	26.81
LOWER NIGER (LOKOJA)	2024/25	224
	2023/24	236
	Five-year wet period	225
	Five-year dry period	137
	Mean	177